Abishek Arumugam Navukkarasan

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Innovation | Research & Development | Design | Hardware | Test Engineering

Work Experience

Lead Engineer **Hudson, MA**

Catalytic Consulting Inc. April 2024 - Present

- Spearheaded the end-to-end design and development of a high-efficiency dual-media PFAS Filtration system from requirements to Pilot-scale Testing.
- Delivered a 30% improvement in breakthrough performance and 10% cost reduction by integrating a novel co-planar external header design that optimized flow dynamics and reduced energy demand.
- Championed CFD analyses to optimize flow architecture, enhancing system reliability and performance.
- Optimized Cycle Test Stand performance for EDI ultra-pure water filtration systems, enhancing reliability and reducing operational costs by over 30% in under 6 months.
- Revamped documentation workflows, facilitating the creation of customer-centric materials, enhancing operational excellence and engineering productivity, and increasing sales.
- Engineered innovative product designs for clients, reducing production costs, boosting revenue, and expanding market presence.

Test Engineering Manager **Westborough, MA**

Amazon Fulfilment Technologies and Robotics *(Contract)* May 2024 - October 2024

- Led a team of 3 engineers to conduct comprehensive testing for an advanced rigid packaging automation machine costing $3M in bridging 3 years in time to market for the customer.
- Delivered actionable performance and functional metrics, including failure root cause through data analysis to product owners, resulting in a 30% reduction in downtime.
- Captured test results, analyzed data, and provided concise, actionable testing reports that translated engineering findings into clear recommendations for project sponsors and stakeholders.
- Developed rigorous test plans, test procedures, and test cases, reducing system fault rates by 20%.
- Conducted Process Failure Mode and Effects Analysis (PFMEA) to proactively identify potential failure modes and optimize the process design and material handling mechanisms based on test findings.
- Created testing checklists and cross-verification matrices to ensure thorough evaluation and data validation.

Test Engineer **Boston, MA**

SparkCharge Inc. April 2023 - December 2023

- Led the development of a critical core product service tool that helped resolve firmware bugs, ensuring brownfield-deployed products remained serviceable, maintaining customer relationships.
- Spearheaded end-to-end development of the service tool - writing PRD, designing 3D mCAD in Solidworks, designing eCAD PCB in Altium, and performing failure analysis, testing, and validation.
- Accomplished unparalleled reliability on the service tool through DFMEA, identifying potential points of failure and implementing corrective actions, which drove customer retention up to 80% and product success.
- Designed a proposal for an End-Of-Line Charger Test Fixture for the in-house manufacturing facility in under 3 months, enabling quicker go-to-market strategies and accelerating the project timeline by 30%.
- Developed and executed test plans for Firmware releases and new component validation towards requirements, and reduced potential product failures by 10%.
- Write, review, or approve documentation like technical specifications, Engineering reports, test specifications, design reviews, and presentations(EVT, DVT, PVT) supporting new product development.
- Run everyday tests in a high-power lab to support R&D design validation prototypes, component validation, EVSE Protocol testing, and analyze data towards applicable ISO, UL, and NEMA Standards.

Development Test Engineer **Chennai, IN**

Agnikul Cosmos Launch Vehicle Pvt. Ltd. January 2019 - August 2019

- Fully realized a semi-cryogenic Test stand from ideation to launch in under 3 months, driving system success and generating crucial data that directly contributed to securing a $3.2 million seed funding round.
- Designed and built 4 test systems to qualify subsystems of a launch vehicle for FAA flight qualification.
- Implemented innovative test design methodologies using principles of design of experiments to strategize and execute efficient testing procedures that generate characterization data to help validate designs.
- Performed Trade-offs for components to balance performance, budget, and reliability.
- Performed analytical work to conceptualize and characterize new designs by assisting with arbitrary numbers related to stress, dynamics, and heat transfer.

EDUCATION

University of Florida **Gainesville, FL**

Master of Science in Mechanical Engineering *2021 - December 2022*

- Team member - Structures and R&D; Gatorloop - Hyperloop team of UF
- Coursework - Innovation 4 Defense, Engineering Innovation, Divergent Thinking, Engineering Project Management, Fundamentals of Production Engineering, Advanced Manufacturing, Geometry of Robots

Anna University **Chennai, IN**

Bachelor of Engineering in Mechanical Engineering *2014 - April 2018*

- Active Participant in Local DIY and Makerspace Communities working with IoT, SBCs, 3D Printing, and AI.
- Coursework - Finite Element Analysis, Fluid Dynamics, Design of Machine Elements, Dynamics of Machines, Principles of Management

PROJECTS

- ***Protection Up High*** *- Dept. of Defense; US Navy; NIWC Atlantic* *August 2022 - December 2022*
  Identified innovative radio DF solutions to help pilots of C-130 aircraft identify ground-based threats
- ***Health monitoring for people in isolation -*** *University of Florida* *May 2022 - August 2022*
  Designed a suite of products for monitoring the physical and mental health of people in Isolation
- ***Semi cryogenic liquid prop engine test stand*** *- Agnikul Cosmos* *April 2019 - August 2019*
  Designed a thrust test stand to qualify a semi-cryogenic liquid prop engine to FAA regulations
- ***SCARA robotic arm 3D Printer*** *- RapidFab 3D Engineering* *December 2017 - May 2018*
  Designed a novel approach to a robotic arm-type 3D Printer
- ***Experimental Lunar Payload*** *- Team Algol/Team Indus* *August 2016 - December 2016*
  Designed a Coke can-sized experimental payload to evaluate the presence of life on the lunar surface

TOOLS & SKILLS

CAD Design: AutoDesk; AutoCad, Inventor, Fusion 360| DS; CATIA, SolidWorks | PTC; Creo

Analysis: Simscale, Ansys| PCBA: Altium

Programming Languages: Python, Embedded C, Matlab

Other skills: GD&T, Machine shop experience, High-Pressure Plumbing, 3D Printing, IoT, ML/DL/AI

Proficient in Microsoft suite, Google suite, Atlassian Tool Suite (JIRA, Confluence), Smartsheet, Linux, and Git

MEMBERSHIPS, AWARDS & CERTIFICATIONS

- Expert Delegate member for ISO 56000 Innovation Management Standards with TC279 at ANSI
- Engineering Innovation Certificate - Engineering Innovation Institute, HWCOE, University of Florida
- Achievement Award scholarship - Herbert Wertheim College of Engineering